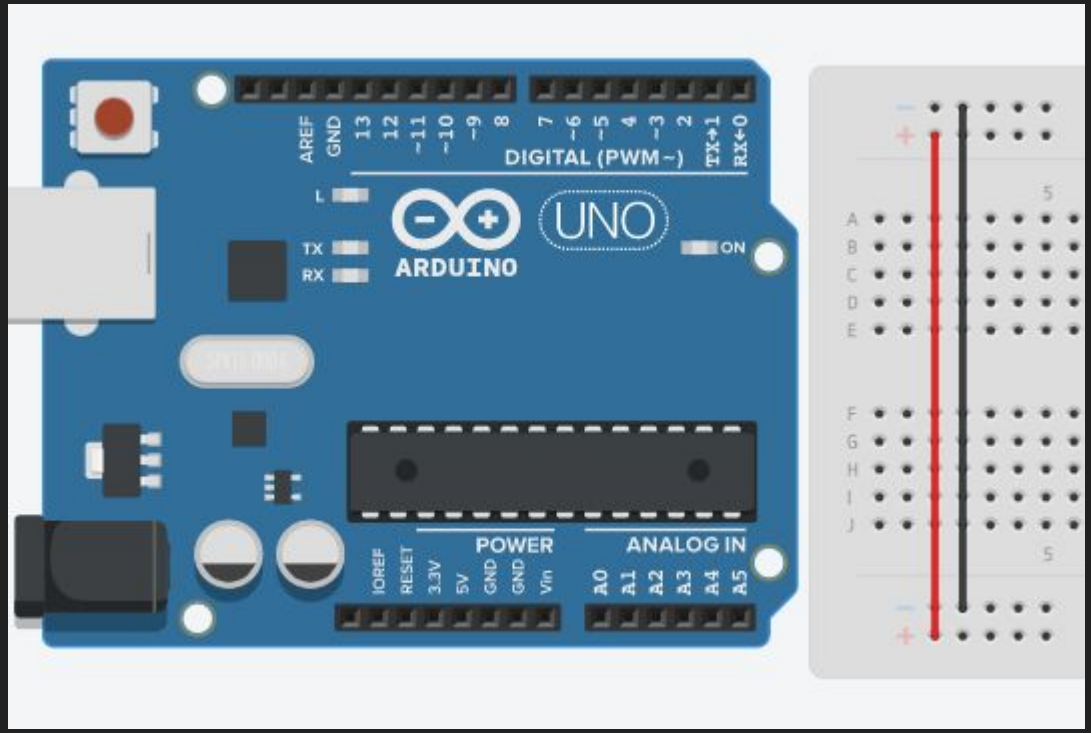
Digital Electronics Interfacing

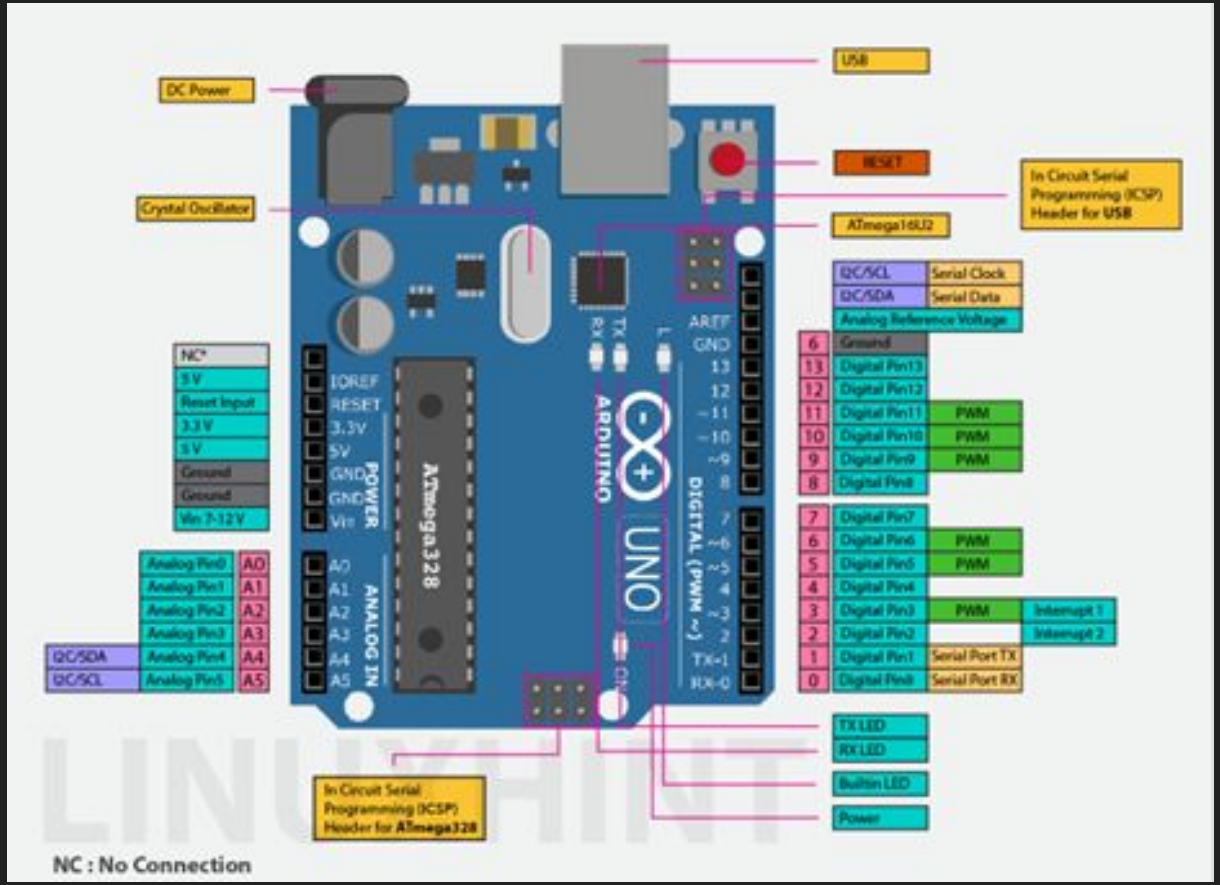
The Arduino



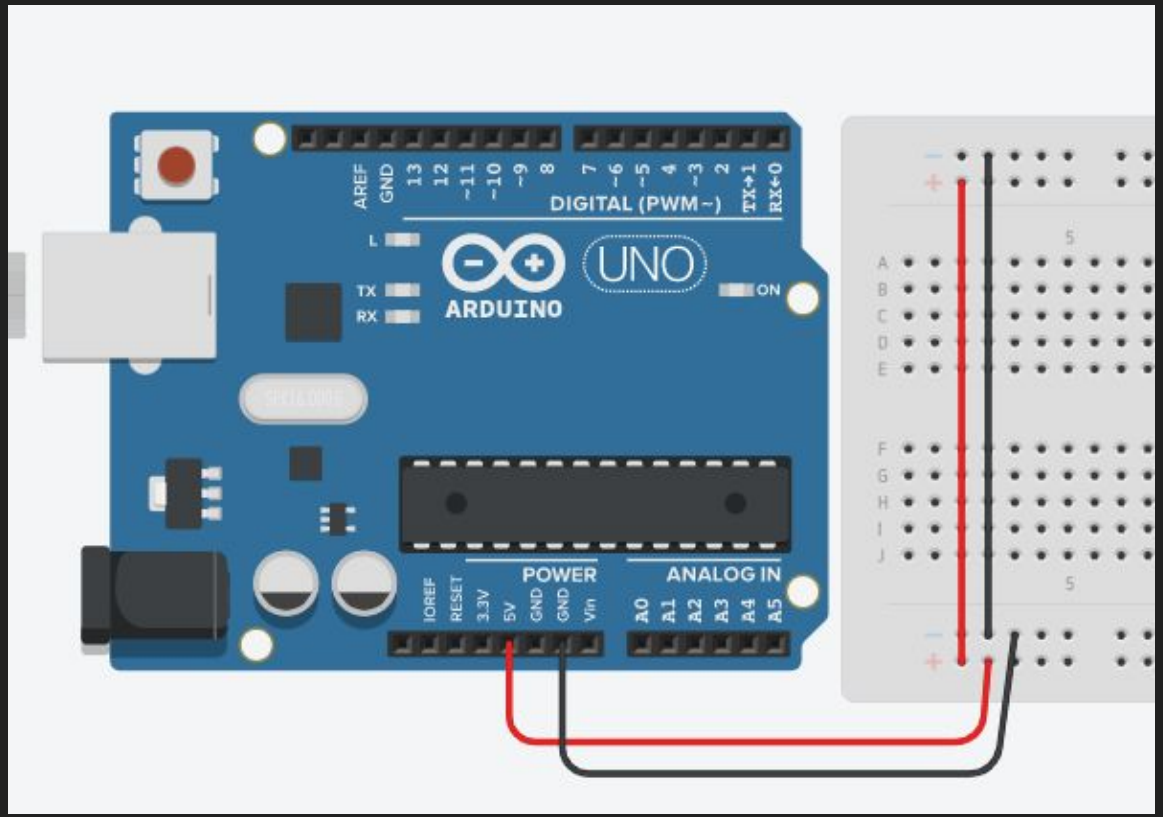
The Arduino-UNO



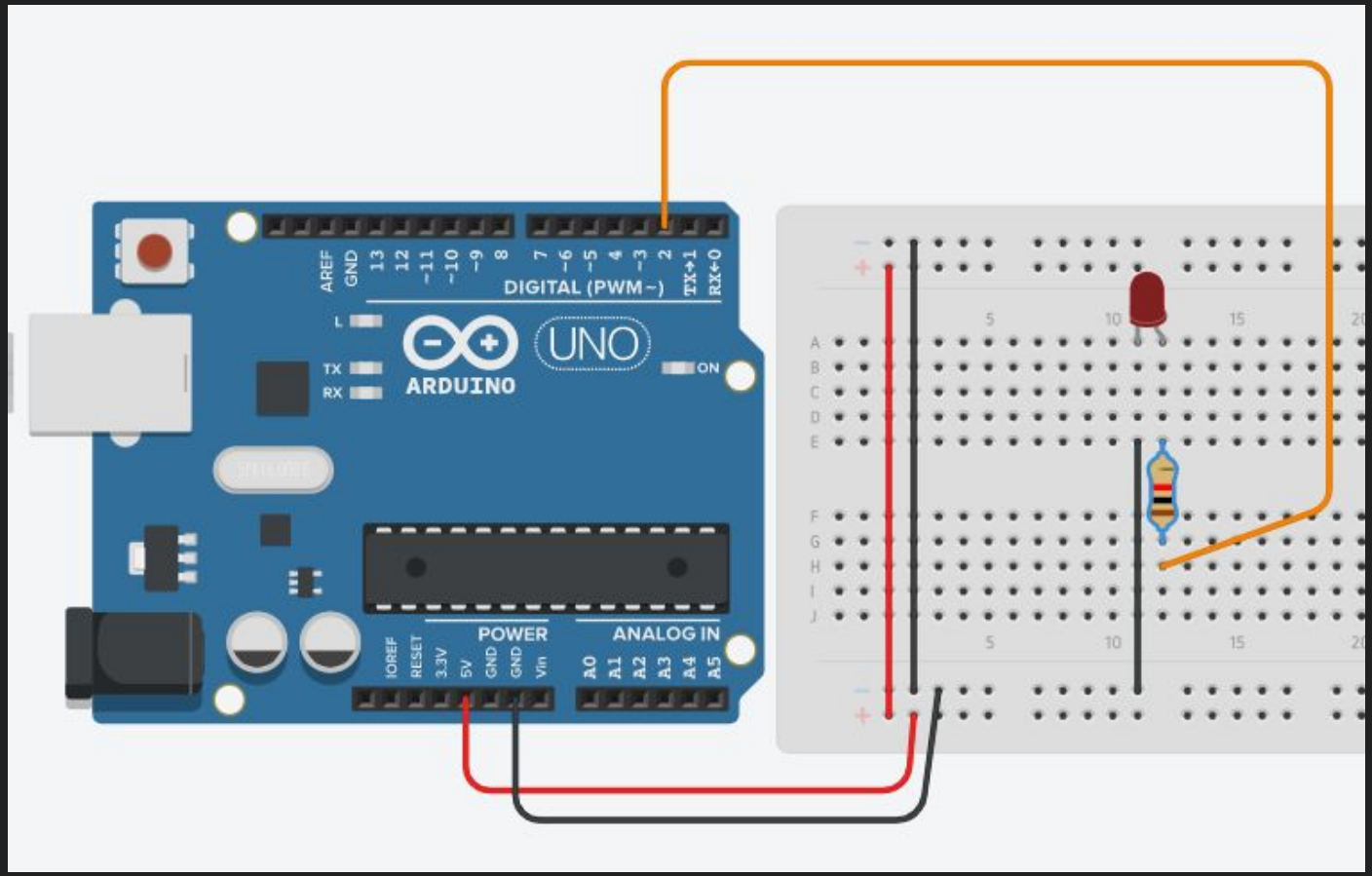
The Arduino-UNO



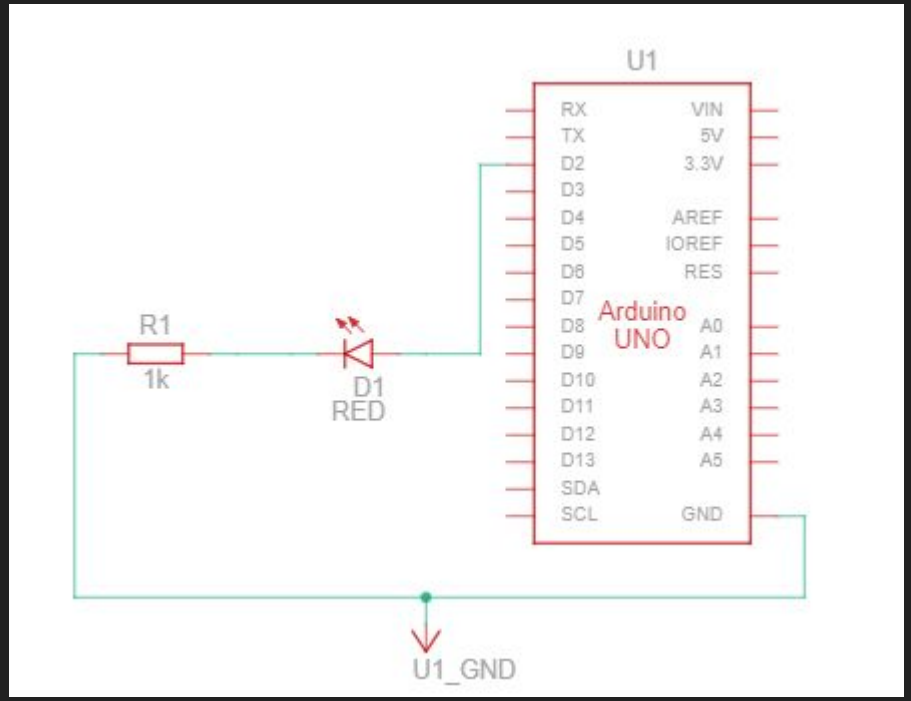
The Arduino-Power and Ground



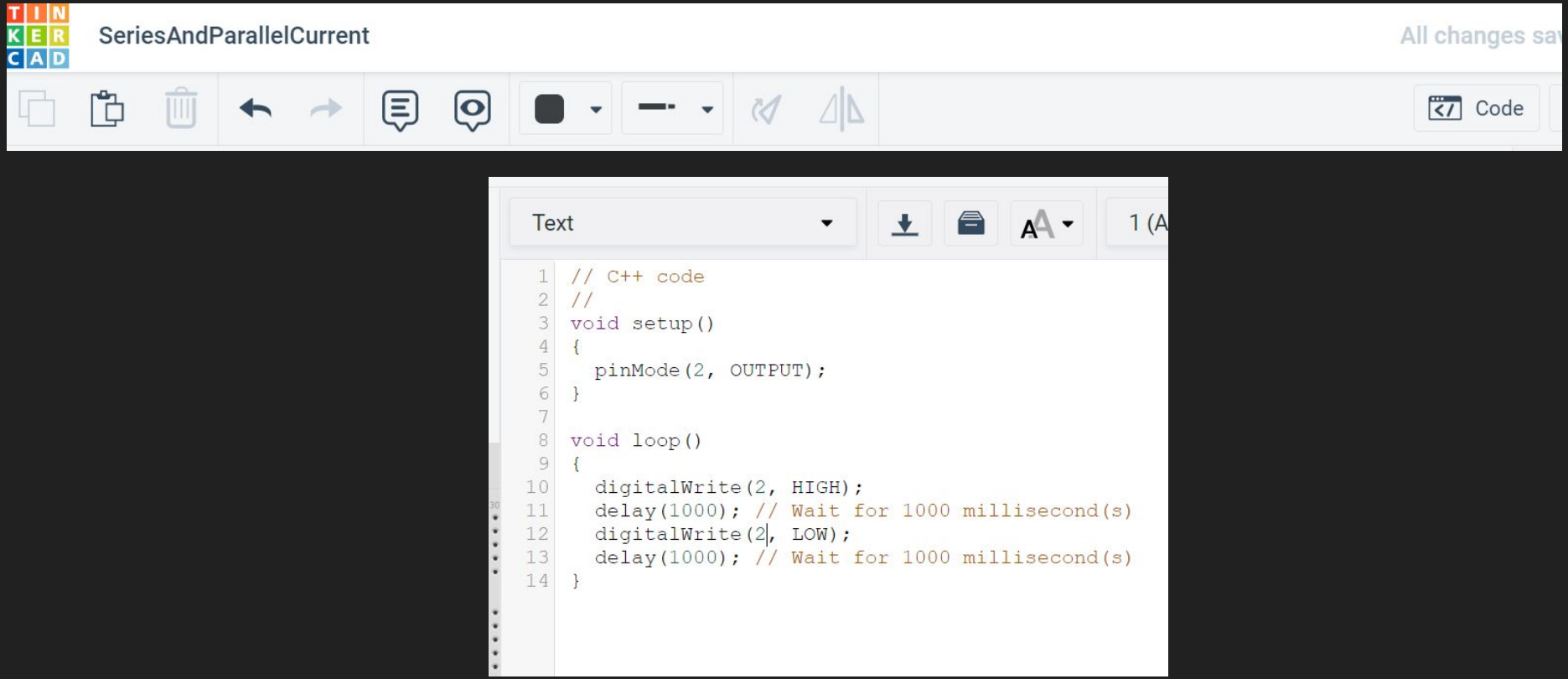
The Arduino-Control IO Pins via Code



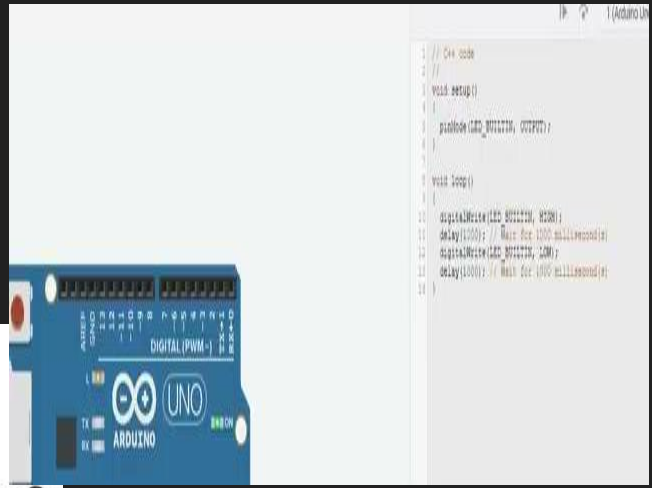
The Arduino-Control IO Pins via Code-Schematic

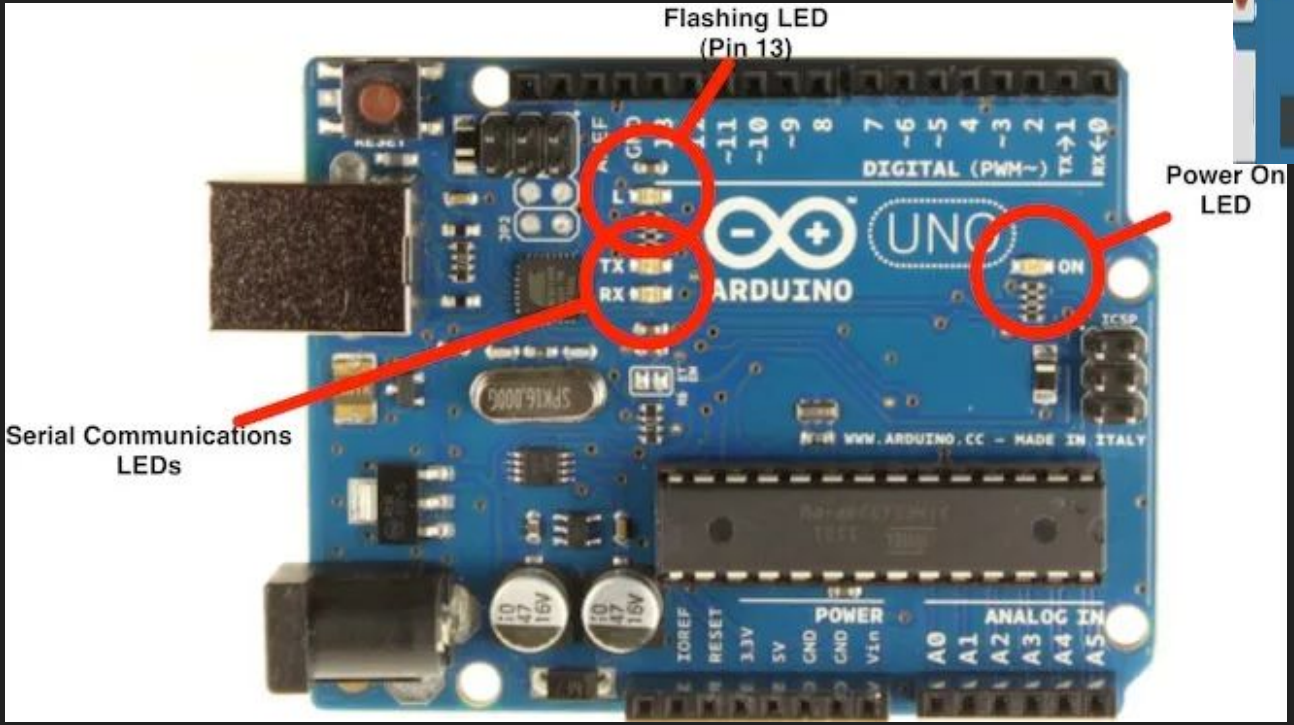


The Arduino-Control IO Pins via Code-Code



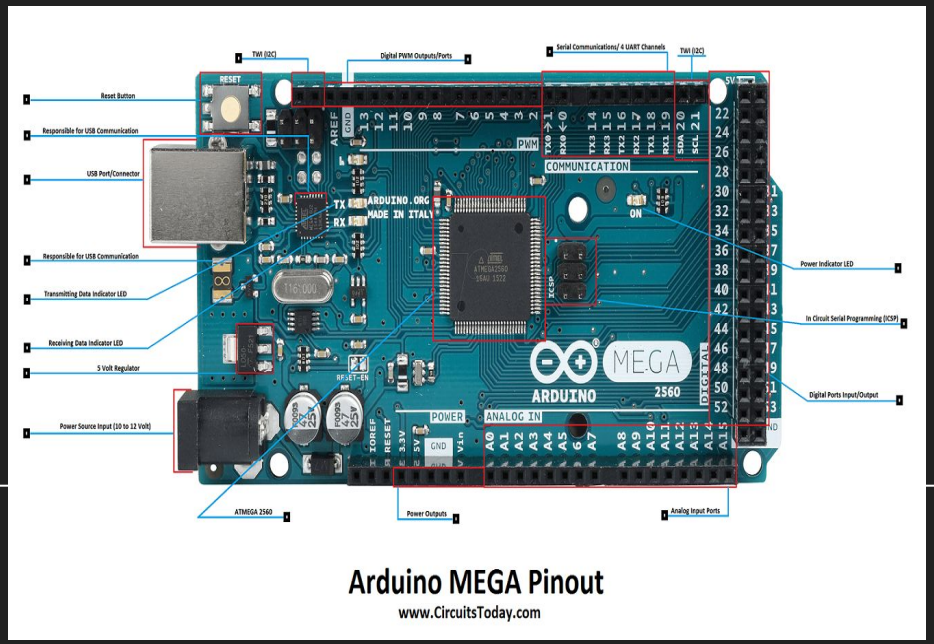
LED_BUILTIN


// C++ code
//
void setup()
{
pinMode(LED_BUILTIN, OUTPUT);
}
void loop()
{
digitalWrite(LED_BUILTIN, HIGH);
delay(1000); // Wait for 1000 millisecond(s)
digitalWrite(LED_BUILTIN, LOW);
delay(1000); // Wait for 1000 millisecond(s)
}
DIGITAL (PWM~)
UNO
ARDUINO


Flashing LED
(Pin 13)
Power On
LED
Serial Communications
LEDs
DIGITAL (PWM~)
UNO
ARDUINO
TX
RX
ON
ICSP
WWW.ARDUINO.CC – MADE IN ITALY
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

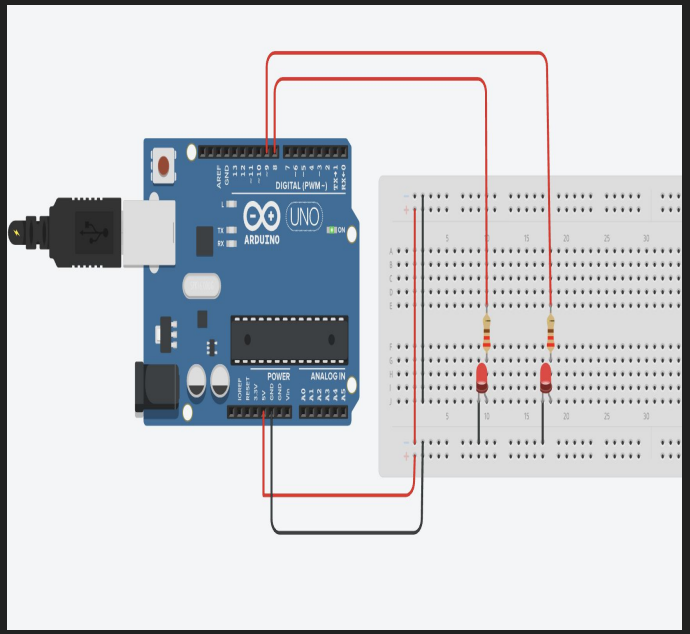
Questions and Exercises

1. What pin numbers are used for Digital IO on the Arduino Uno?**D0 to D13**.
2. What pins can be used for Analog IO on the Arduino Uno?**A0–A5**
3. What pins supply 5V on the Arduino Uno? **5V pin supplies 5 volts on the Arduino Uno**
4. What pins are used for a connection to ground on the Arduino Uno? GND
5. Find and embed a pinout for the Arduino Mega.



Questions and Exercises

6. Create an Arduino circuit that turns on 2 LEDs simultaneously on and off every 2 seconds. Embed a screenshot of the live circuit and the code below:




```
int led1 = 8;
int led2 = 9;

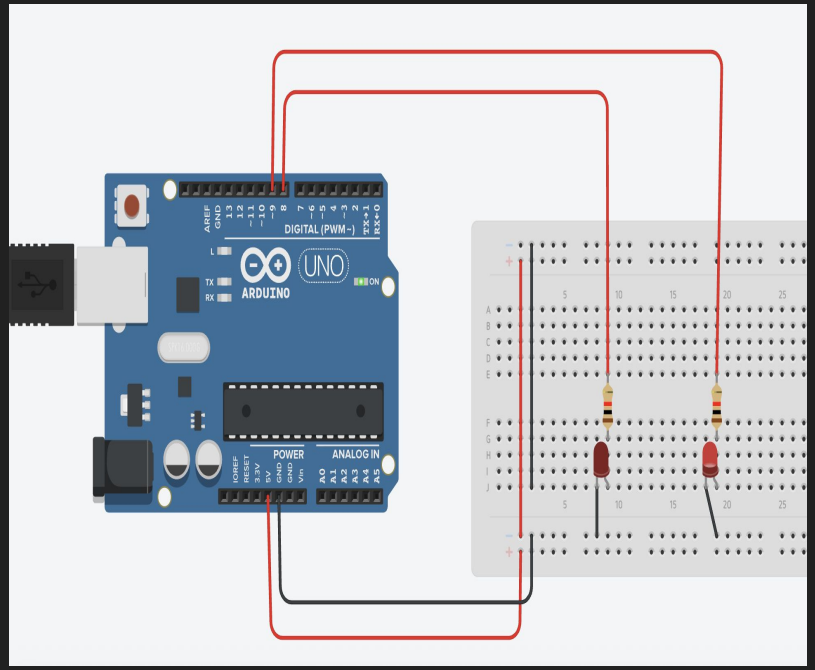
void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  // Turn both LEDs ON
  digitalWrite(led1, HIGH);
  digitalWrite(led2, HIGH);
  delay(2000);   // wait 2 seconds

  // Turn both LEDs OFF
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
  delay(2000);   // wait 2 seconds
}
```

Questions and Exercises

7. Create an Arduino circuit that turns on 2 LEDs in alternating order on and off every 2 seconds. When one light is on the other is off. Embed a screenshot of the live circuit and the code below:




```
int led1 = 8;
int led2 = 9;

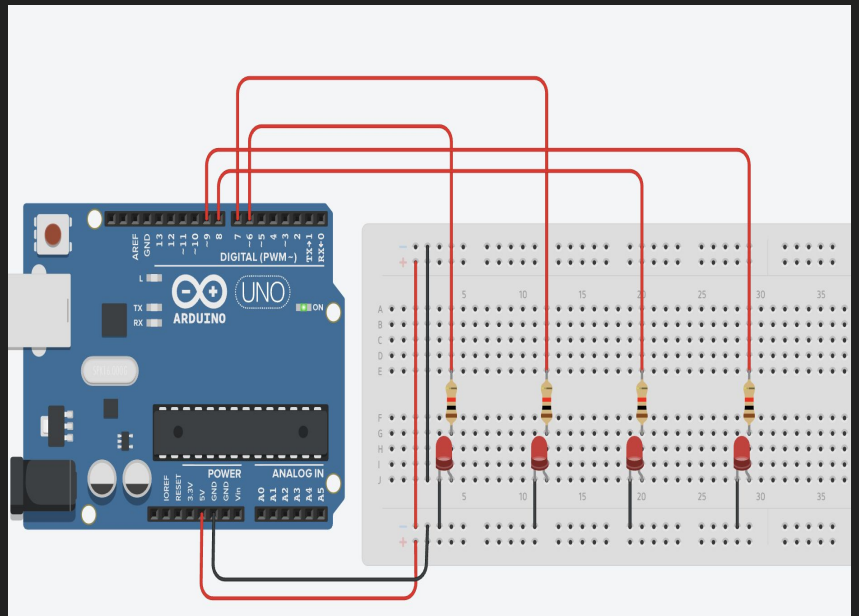
void setup() {
  // Set both pins as output
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  // Turn LED1 ON and LED2 OFF
  digitalWrite(led1, HIGH);
  digitalWrite(led2, LOW);
  delay(2000);   // wait 2 seconds

  // Turn LED1 OFF and LED2 ON
  digitalWrite(led1, LOW);
  digitalWrite(led2, HIGH);
  delay(2000);   // wait 2 seconds
}
```

Questions and Exercises

8. Create an Arduino circuit that turns on 4 LEDs one after another with a 1 second delay in between. After the 4th LED turns on for 1 second all LEDs turn off the cycle repeats. Embed a screenshot of the live circuit and the code below:




```
int led1 = 6;
int led2 = 7;
int led3 = 8;
int led4 = 9;

void setup() {
  // Set all LED pins as outputs
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(led3, OUTPUT);
  pinMode(led4, OUTPUT);
}

void loop() {
  // Turn LEDs on one by one
  digitalWrite(led1, HIGH);
  delay(1000);

  digitalWrite(led2, HIGH);
  delay(1000);

  digitalWrite(led3, HIGH);
  delay(1000);

  digitalWrite(led4, HIGH);
  delay(1000);

  // Turn all LEDs off
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
  digitalWrite(led3, LOW);
  digitalWrite(led4, LOW);
}
```

Assignment TI /10

Create an Arduino circuit that has 6 LEDs in a circle. Turn the lights on, one after the other in clockwise order. Wait 1 second in between each light. Then turn the lights off one after the other with a 1 second delay in between. Then repeat but this time the delay is 500ms. Finally repeat one last time with a delay of 250ms. Embed a screenshot of the live circuit and the code below:

https://www.tinkercad.com/things/iFOkTeqGy7M-daring-tumelo

